

Automated detailing of exterior walls using NADIA: Natural-language-based architectural detailing through interaction with AI

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ABSTRACT

Learning building information modeling (BIM) systems has always been a challenge for BIM adoption. Although groundbreaking performances of large language models (LLMs) have inspired many researchers to consider an LLM as a potential BIM control method using natural language, a specific method of utilizing LLMs for automated BIM model detailing has not yet been proposed. This paper proposes an LLM-BIM chaining framework to enable architectural design detailing using natural language, instead of using menu-based user interfaces, named “Natural-language-based Architectural Detailing through Interaction with AI (NADIA)”. The NADIA framework is based on three main approaches: 1) separating the specification of the wall layers from the creation of the wall layers; 2) appropriate instruction prompting to guide the LLM to minimize irrational responses and produce engineering rational details; and 3) LLM-BIM chaining to seamlessly link a BIM authoring tool and an LLM. The effectiveness of NADIA was validated based on two main aspects: its accuracy in generating details that adhere to specified design requirements from users—as a design assistant—and its compliance with general engineering requirements—as a design consultant. The validation was achieved through tasks that involved generating 240 and 1,920 exterior wall details, respectively. NADIA achieved an average accuracy of 83.33% in generating logically coherent details in line with the required design conditions. For thermal performance requirements, it demonstrated a mean accuracy of 98.54% in complying with the American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) 20.1–2019 standard. Despite being in its early stages, NADIA’s potential for developing and refining architectural details through natural language-based interactions between architects and machines is promising.

1. Introduction

Building information modeling (BIM) has evolved to enable architects to conduct a comparative analysis of design options [1], accurate cost estimation [2], automated code checking [3,4], clash detection [5], safety management [6], and construction simulation [7,8]. However, to effectively use BIM software, one must learn the system structure, commands, and user interfaces to perform such tasks [9]. Such a ‘need to educate professionals about BIM’ has been identified as one of the major hurdles to BIM adoption [10,11].

The recent integration with generative artificial intelligence (AI) technology exhibits promising opportunities in how architectural design software can be utilized and operated [12]. Particularly, generative adversarial networks (GANs) [13] and generative language-image (i.e., multimodal) models, such as DALL·E 2 [14], Stable Diffusion [15], and

Midjourney [16], have shown potential for assisting architects in exploring early design alternatives through design image generation [17–21], floorplan generation [22–24], and building performance prediction [25,26]. In particular, the technology for creating schematic images using natural language has reached a commercial level through image-diffusion-based approaches, such as Veras [27] and SketchUp Diffusion [28] which integrate directly with BIM tools. The use of large language models (LLMs) for schematic design generation has become commonplace at work and school due to significantly reduced rendering time and improved quality with simple natural language prompts. However, LLM-based automated detailing has not been extensively explored, although many may have considered the possibility of using an LLM for detailing. One challenge is to control LLMs to produce “engineering-wise rational” results based on arithmetic and symbolic reasoning [29]. Moreover, unlike schematic design, BIM-based design

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detailing requires not only consideration of general domain knowledge of architectural design but also an understanding of multi-hierarchical, object-oriented programming (OOP) data structures employed in BIM software [30]. On the other hand, detailing, which takes the longest time during a design process and requires complex BIM operations, is an area where architects can greatly benefit from natural language-based interactions rather than command- and button-based interfaces.

To enable architects to develop customized design details using natural language, this study proposes a framework named “Natural language-based architectural detailing through interaction with AI (NADIA)” which integrates LLM and BIM authoring software. The NADIA framework takes three key steps: 1) Decompose the wall design tasks: Instead of designing the entire wall at once, it separates the description of each layer from the actual creation of the wall. 2) Guide the AI for better results: It uses specific prompts to guide the LLM to provide logical engineering details and avoid nonsensical responses. 3) Connecting AI and design software: Seamlessly connect the LLM to a BIM tool, allowing them to work together seamlessly.

The NADIA framework was validated from two aspects: The first aspect aimed to assess NADIA’s performance as a “design assistant,” focusing on attesting the framework’s ability to generate exterior wall details that adhere to an architect’s explicit requirements, considering the initial motivation to assist architects in design detailing through natural language-based interactions. The second aspect focused on evaluating NADIA’s performance as a “design consultant,” which involves generating design solutions that comply with engineering standards, such as thermal performance, even when certain design criteria or guidelines are not directly given.

The scope of this paper was limited to the generation of exterior wall details. The exterior wall was chosen as the primary focus due to its prevalence as the most commonly used object type in a BIM model [31]. Among wall objects, exterior walls require detailed specifications based on various factors including climate, building type, materials, and structural frameworks [32–35]. This multi-faceted nature of exterior wall detailing makes it an ideal subject for validating NADIA’s ability to generate intricate design details using natural language.

This paper is organized into seven sections. Section two reviews previous studies relevant to automated design generation and the use of LLMs in the architect, engineering, and construction (AEC) industry. Section three introduces the NADIA framework and how it is implemented in this study. Section four outlines the design for validation of the NADIA framework. Section five analyzes the results of the validation, and section six discusses the results. Finally, section seven concludes the paper, presenting the contributions, limitations, and suggestions for future research.

2. Previous studies

The vision of design development through conversations between an architect and a computer in natural language such as “human–computer symbiosis” [36] or “clerk” [37] was proposed in the late 1950s and early 1960s. It has come closer in the conceptual and schematic design generation thanks to the rapid advances in generative artificial intelligence (AI). However, the interactive development of details through natural language conversations between an architect and a computer remains unexplored. Computer-aided design (CAD) and BIM have been developed to enhance productivity and decision-making. Consequently, pertinent studies have been conducted to automatically generate and optimize the design alternatives. A prevailing research topic was the use of parametric modeling approaches to embed expert knowledge into a system as mathematical rules, and subsequently adjust the parameter values to create multiple design alternatives [38,39]. Earlier studies focused on developing parametric rules that can represent crucial features of building objects within BIM environments. For example, Sacks et al. [40] developed a template to demonstrate the feasibility of automated structural design using parametric rules. Lee et al. [41] proposed

a method to specify design and engineering knowledge as building objects and behaviors using parametric modeling. Parametric-model-based optimization was another research topic in the area. Liu et al. [42] introduced a rule-based automation method to minimize material waste in light-frame residential buildings. Mangal and Cheng [43] proposed a BIM-based framework using a hybrid genetic algorithm to automate steel reinforcement optimization in reinforced concrete frames. Tafrout et al. [44] and Bourahla et al. [45] developed approaches to optimize the structural performances of architectural configurations. Numerous studies have explored design generation and optimization through parametric modeling, but common limitations are evident. Parametric modeling has a high entry barrier because it requires an understanding of both architectural knowledge and complex relationships between geometrical elements. Moreover, it is known to be labor-intensive and time-consuming to identify and define a large number of input parameters and their relationships [38]. Further, accommodating new cases necessitates modifications of algorithms [46].

The emergence of image-generation AI models such as generative adversarial networks (GANs) has provoked novel approaches to generating design alternatives [13]. For example, Zhang et al. [18] proposed a stylistic scene enhancement of three-dimensional indoor images using conditional Wasserstein GAN. Sun et al. [17] trained a GAN using a dataset of façade and label information from a historical project and proposed its use in historical urban renovation. Qian et al. [19] employed a self-sparse GAN to generate architectural shape sketches for early-stage design. Luo and Huang [22] introduced a GAN-based model that combines vector generation and raster discrimination for floor plan generation. More recent AI models, which are trained upon the fusion of information from text and images, allow users to interactively generate images based on prompts. Ploennigs and Berger’s study [47] performed a comparative analysis of the design outcomes from interior and exterior design visualization tasks using DALL-E 2, Stable Diffusion, and Mid-journey. However, the use of such AI models still innates common limitations with GANs-based approaches. While these AI models enable the interactive development of architectural images in line with the architects’ prompts, their use is still confined to the preliminary design phase, restricted to image-based assignments [47]. The outputs of these models are solely images, lacking detailed information about the building projects and absent of their subordinate building components and properties.

To address these inherent limitations, this study proposes a framework that integrates BIM with an LLM to improve design automation flexibility through interaction with architects. Recently emerging LLMs, including OpenAI’s generative pretrained transformer (GPT), Google’s pathways language model (PaLM), which is the basis for Bard, and Meta’s large language model meta AI (LLaMa) [48], are known for their comprehensive performance in natural language processing (NLP) tasks like text generation, question and answering, summarization, and translation [49]. Previous studies in the AEC industry also have integrated language models in BIM systems for querying data, leveraging their ability to process both computer-readable and human languages. Lin et al. [50] developed a question-and-answering system for information search related to BIM topics using bidirectional encoder representations from transformers (BERT). Similarly, Wang et al. [51] combined a BERT model and case-based reasoning to improve information retrieval from the BIM model. Zheng and Fischer [52] developed a framework that integrated GPT and BIM databases for natural language-based information retrieval. Nevertheless, the scope of these studies was limited to language-model-based data retrieval from a BIM model. The concept of automated detailing, integrating an LLM and a BIM authoring process, has not yet been explored.

A key challenge of LLMs to be integrated with BIM systems for design detailing lies in the absence of a generalized method capable of consistently returning rational design details. To address this challenge, the NADIA framework employed “LLM chaining” which involves segregating BIM-based design detailing tasks into manageable subtasks,

each guided by specific instruction prompts. This approach is based on the idea of leading the LLM through a series of steps, similar to how a human expert would approach problem-solving, enabling LLMs to handle complex tasks [53]. Furthermore, the NADIA framework enhances the consistency of the results through instruction prompting, by clearly defining the subtasks and expected return [54]. This includes specifying the properties and values essential for representing building components within BIM environments [55]. Through the strategic division of design tasks, the application of instruction prompting, and the synthesis of these elements via LLM chaining, the NADIA framework aims to enable architects with an interactive, flexible, and controlled solution for automating architectural design detailing within the BIM environment.

3. NADIA framework

3.1. Design detailing using NADIA

The NADIA framework builds upon the generalized iterative process of design elaboration [56], with a specific emphasis on BIM-based detailing processes [30,57]. Fig. 1 illustrates the iterative process of design detailing in BIM authoring, showcasing the evolution from an initial conceptual model to a more detailed one. The basic BIM model is a digital representation containing the core components of the building project, yet it requires further detailing for completion. The detailing phase begins by preparing a design detail for a specific component type. This detail may require iterative modification to align with design standards or engineering requirements. The enhanced detail is integrated into the BIM model through a three-step process: first, by retrieving relevant objects that need their details updated; second, by instantiating the new details within the model; and third, by replacing the objects' original details with the updated details. This process is repeated for all building components requiring improvement, resulting in a comprehensive and detailed BIM model.

Building on the concept of design detailing in BIM authoring, the NADIA framework leverages a three-step natural language interaction between an architect and a BIM authoring tool for design detailing: 1) generate a customized detail class with a high level of development (LOD); 2) iteratively modify the detail class until the architect is satisfied; 3) instantiate the detail class and replace the existing low LOD object with the newly created high LOD detail instance. To support this process, the NADIA framework comprises three primary modules: the architect-NADIA interface module (interface module in short), the task execution module, and the design implementation module, as depicted in Fig. 2. The interface module allows an architect to describe design detailing tasks through conversational input via a front-end medium. In

the backend, the task execution and design implementation modules collaborate to interpret the user prompts and apply the design intent into machine-readable data, subsequently representing it within the BIM environment.

3.1.1. Detail generation

As a first step, architects can initiate an interaction by typing a task prompt through the interface module or by using a speech-to-text module that comes standard on most computer operating systems. An example of an input task prompt is as follows:

“Propose an exterior wall detail for a housing project located in Poissy, France. The finishing must be durable, white-colored material for the exterior, and light-colored wooden material for the interior.”

This task prompt clarifies the user's design task, such as the generation of an exterior wall detail, and outlines conditions related to the target, including building occupancy (e.g., housing project), region (e.g., Poissy, France), and material usage (e.g., durable, white-colored material for exterior). While the request may be clear, neither separate use of LLM-based input-output prompting nor conventional BIM authoring can automatically generate the wall detail as explicitly described within the task prompt. Fig. 3 illustrates how a considerable portion of both approaches still requires manual intervention, while also depicting how the proposed approach can resolve this issue. Through input-output prompting use of LLM, the input task prompt would return a process or descriptions related to what was requested by the user as shown in Fig. 3A, which requires an extra elaboration of the design details based on the answered information and further implementing them within the BIM environment. In contrast, conventional BIM practices require architects to input all the necessary information to generate a wall detail that satisfies the given conditions, as shown in Fig. 3B. This may encompass detailed specifications of dependent elements and attributes, such as the required thickness and thermal performance of each layer, and the inclusion of other layers, such as insulation or waterproofing, even if the detailed information were not initially specified.

The interface module streamlines such design detailing tasks by conveying the architect's design prompt to both the task execution and design implementation modules. It facilitates end-to-end chaining, ensuring the delivery of necessary details generated by these modules into the BIM environment, as depicted in Fig. 3C.

The task execution module processes the design task requested by the architect. The module utilizes instruction prompting, which specifies the expected return format of the interim output, thereby streamlining the LLM chaining process. Specifically, LLMs are deployed to two segregated subtasks: task type classification and designed result generation. In the first step, the task execution module analyzes and classifies the task

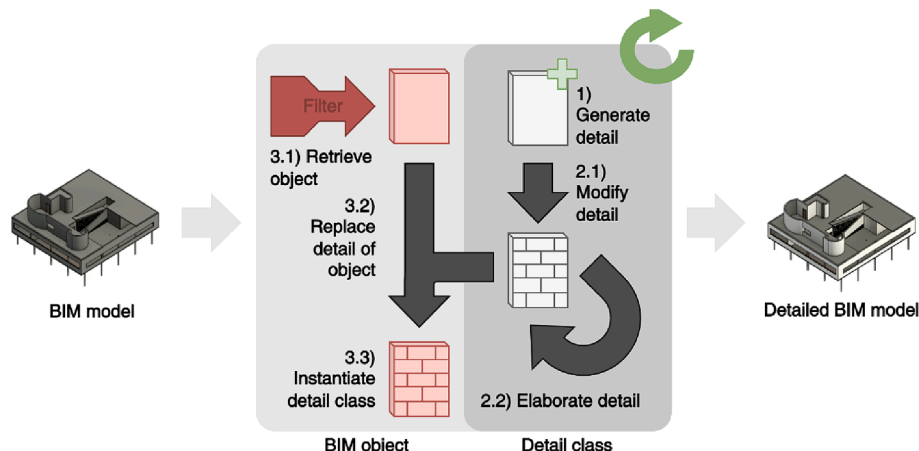


Fig. 1. Design Detailing in BIM Authoring.

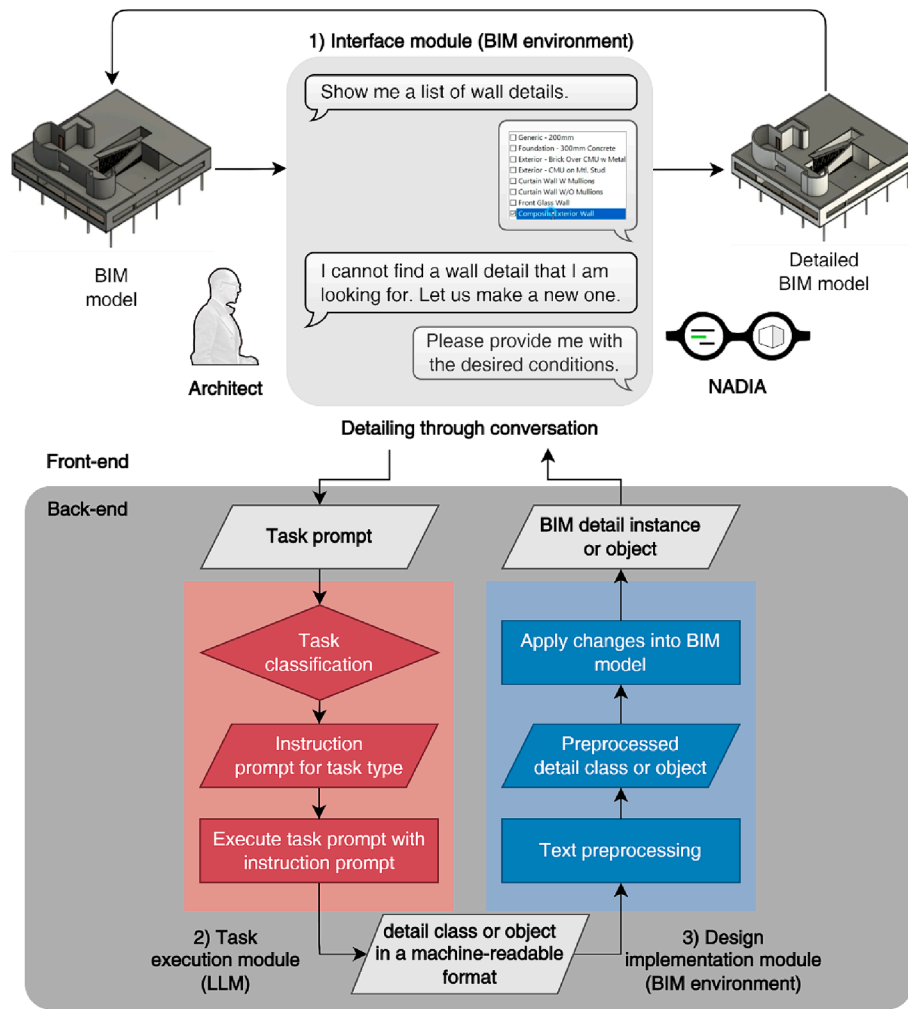


Fig. 2. NADIA framework.

prompt by activity type (e.g., create, modify, retrieve, and delete), component type (e.g., walls, windows, columns, and others), and target levels (determining whether the task is related to detail classes or BIM objects) under the given instruction prompt (as shown in the first 'Instruction prompt' of Fig. 3C) to clarify the design task. These instructions include a specified data exchange format like JavaScript Object Notation (JSON) or markup languages, outlining each value of activity type, component type, and level to collectively identify the task type. This combination of components forms a task type; for instance, "Generate a wall detail class" (refer to the first 'Interim output' of Fig. 3C). In the second step, the role of the LLM can be seen as a text generator that assigns appropriate properties which are required to accurately represent the required design details. Based on the task that is identified from the first step, the module appends appropriate instruction prompts to generate a desired design output. These instructions also specify a data exchange format and further outline the details required to represent the design solution in an OOP-based BIM environment. For example, to generate a wall detail class, the instructions should include the name of the wall detail (i.e., property), the layers included in the wall detail, and the thicknesses, materials, thermal conductivities, and types of each layer (i.e., dependent objects and their properties), as depicted in the second 'Instruction prompt' in Fig. 3C. The flexibility of this structure enables compatibility for generalizing methods across other building component types (e.g., windows, doors, and slabs). The module then executes the task prompt along with the instruction prompts and returns a detailed class in textual format (refer to the second 'Interim output' of Fig. 3C).

The design implementation module receives the data generated by the task execution module and incorporates it within the BIM environment. This module initially preprocesses—cleans and normalizes—the names and values of entities and properties in the automatically generated design detail data to align with the naming conventions and internal structure of the classes predefined within BIM software. After preprocessing, the detail classes are loaded into algorithms that create a design detail instance in the BIM environment. For example, this process can generate a wall detail, as shown in the 'Output' of Fig. 3B from the given task prompt.

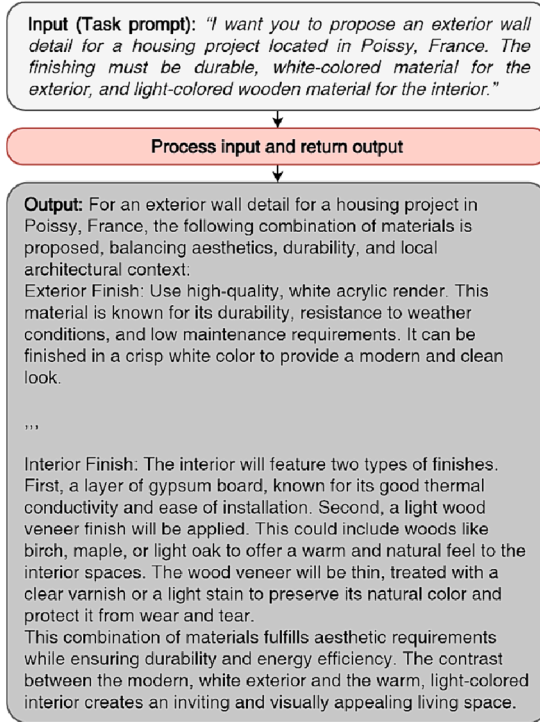
3.1.2. Detail modification

If the task prompt is classified as a modification of a detail class, the target detail class is retrieved and incorporated into the process, as depicted in Fig. 4. A modification task prompt can be entered into NADIA as follows:

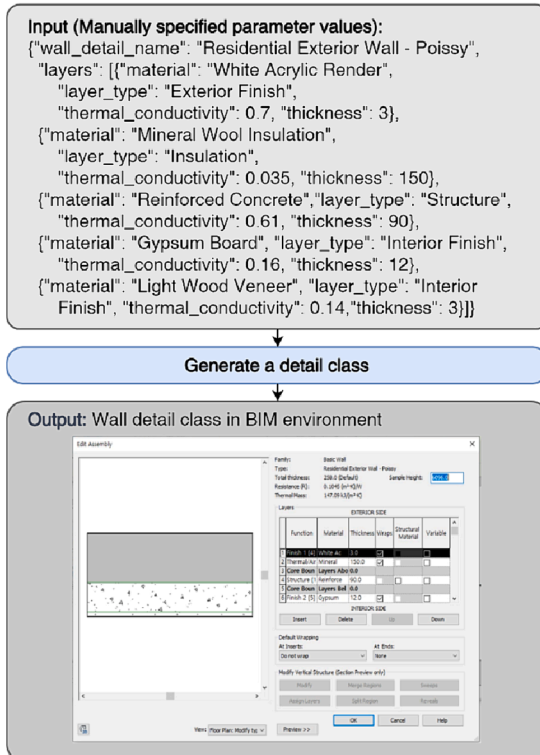
"Can you adjust the wall detail? It needs a membrane layer and replacement of the structural layer to be brick."

If a detail is already selected for modification, the target can be determined automatically. Otherwise, based on the 'component_type' and 'level' identified from the task classification, possible target details are queried for retrieval. Then, from the list, the user can select the target detail to modify. The design implementation module presents the retrieved target detail class in JSON-formatted text. This representation is then combined with the task and instruction prompts. The task execution module receives this combined input and implements the

A Input-output prompting (LLM)



B Conventional BIM authoring (BIM)



C NADIA

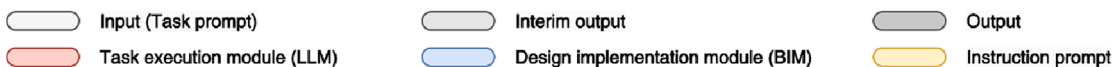
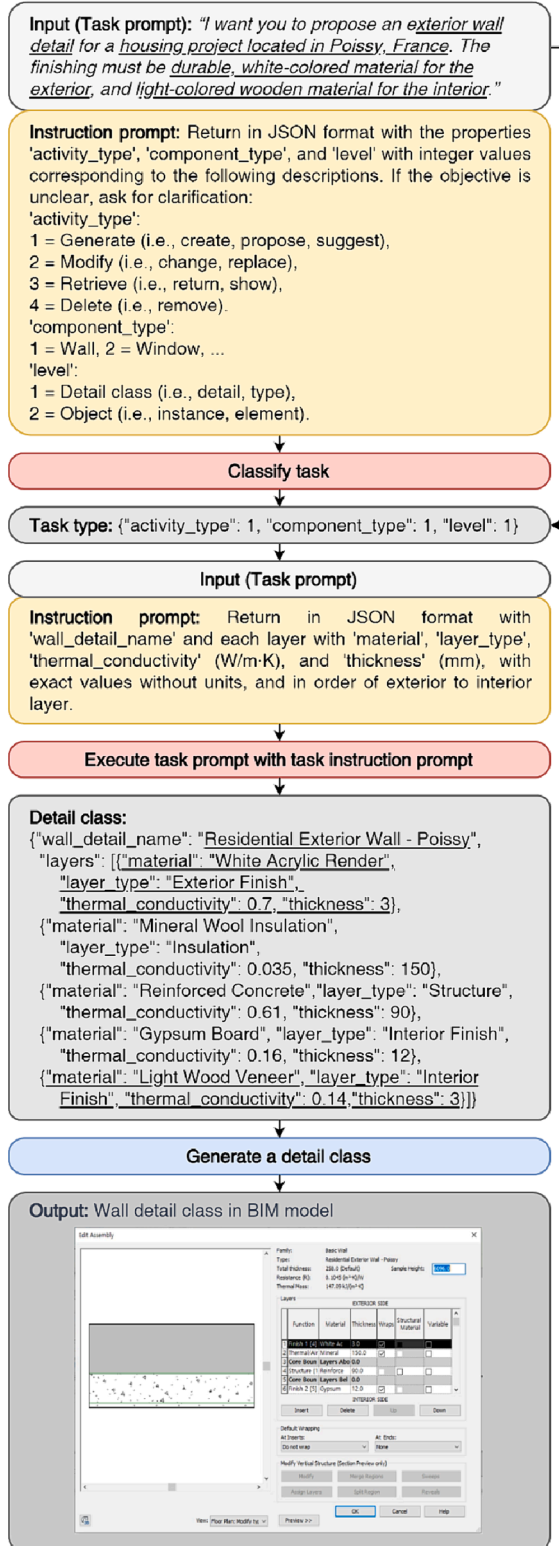


Fig. 3. Examples of wall detail generation using input-output prompting, conventional BIM authoring, and NADIA framework.

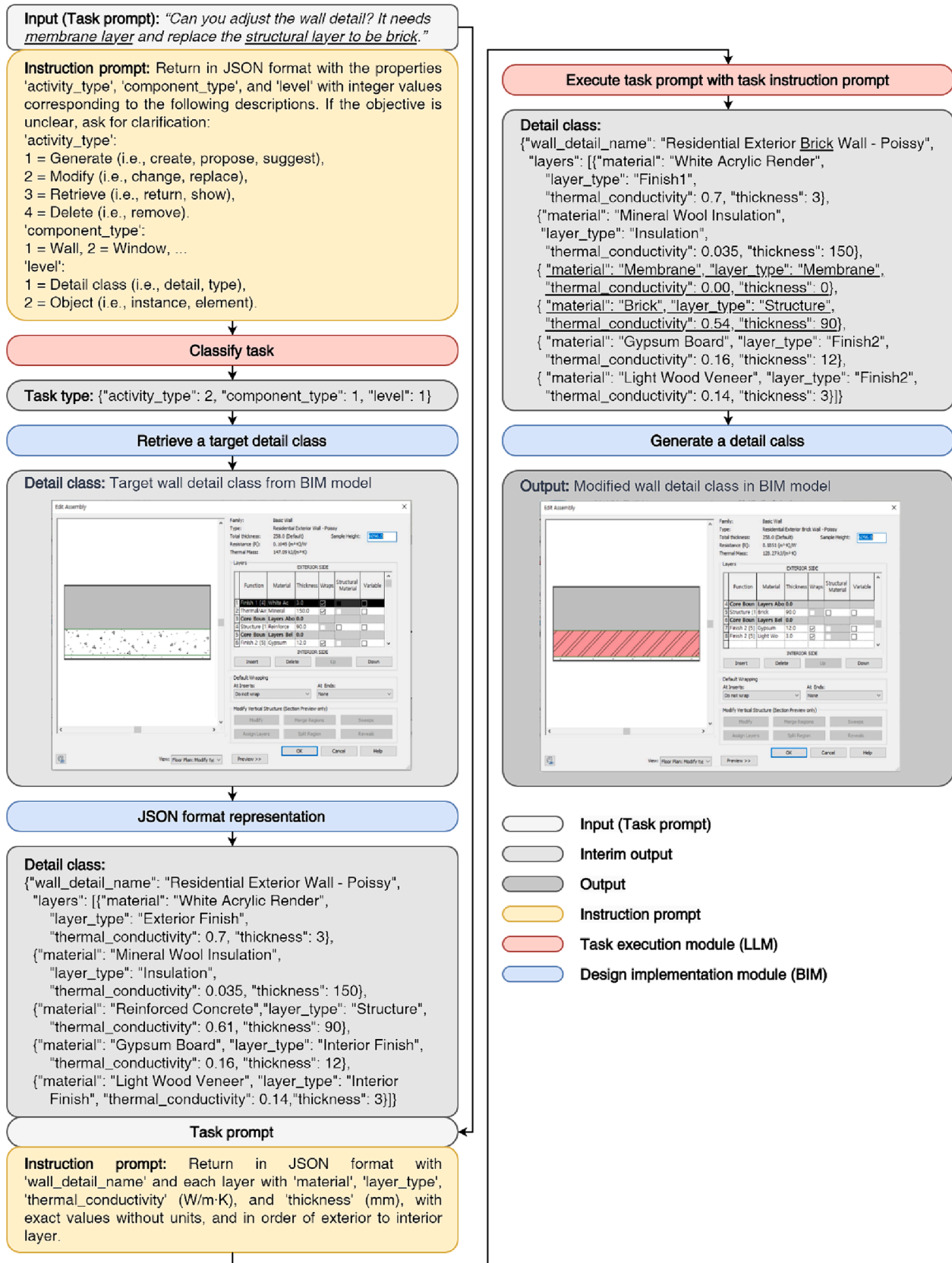


Fig. 4. Example of wall detail modification using NADIA framework.

necessary modifications, returning the result in the same JSON structure. It is re-entered into the design implementation module to generate a new wall detail class with the required modifications applied. If the resultant detail class satisfies the architect, it can then be used to detail the objects in the BIM model.

3.1.3. Object replacement with the newly generated detail

For replacing the details of objects, the following task prompt can be entered (Fig. 5):

“Can you update the detail of all exterior walls with the one we just generated?”

To identify the objects requiring modification, the design implementation module retrieves all objects within the specified ‘component_type,’ extracting each object along with their identifiers (IDs) and

properties in JSON format. For example, determining whether a wall object is an exterior wall may depend on properties such as the names of the adjacent rooms [58]. This extracted JSON is then merged with the task and instruction prompts. The task execution module processes this combined input by first filtering the target objects based on their IDs, then updating specific properties and values that require modification, such as the details employed. The resulting output is then sent back to the design implementation module, which then updates the details of the target objects accordingly. This process completes the three-step interaction process for design detailing using NADIA.

3.2. Implementation

Although alternative products, such as ArchiCAD for the BIM environment and LLaMa for the LLM, could substitute for our chosen tools,

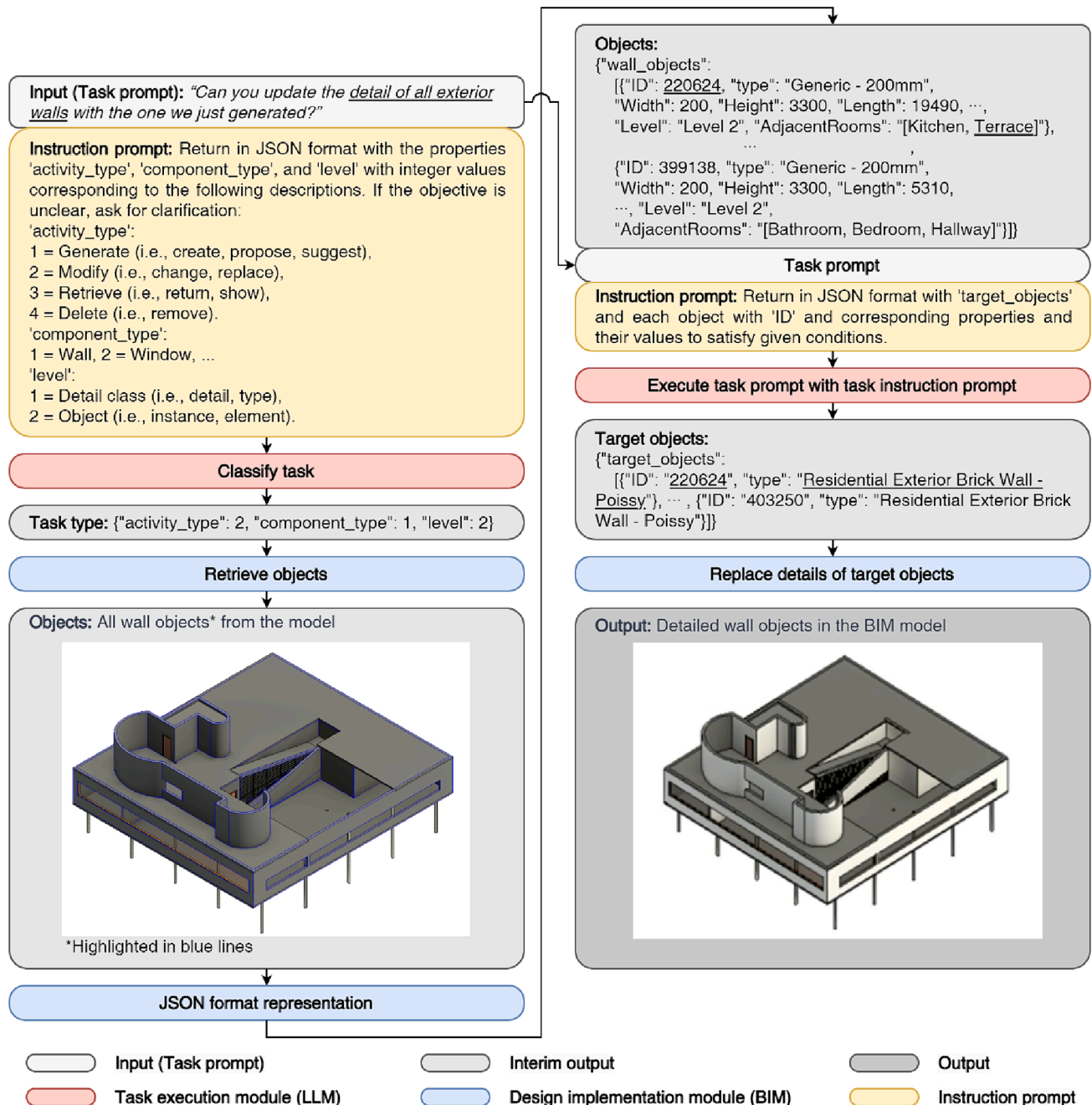


Fig. 5. Example of object replacement with new wall detail using NADIA framework.

we employed GPT-3.5-turbo-0301 as the LLM and Revit 2024 as the BIM authoring software in implementing and validating the NADIA framework.

The interface, task execution, and design implementation modules were integrally implemented as add-ons to Revit, using its C# application programming interface (API) based within the .NET framework development environment. Although GPT-3.5-turbo-0301 did not require fine-tuning for isolated NLP tasks, including text comprehension, classification, or generating JSON-structured results, its integration into the BIM system necessitated the design of specific transactions and system chains [59]. Despite the availability of libraries such as LangChain [60] or Semantic Kernel [61], we employed hard-coded for the chaining. The comprehensive data flow of the NADIA framework is

depicted using a unified modeling language (UML) sequence diagram in Fig. 6. Initially, the user-entered task prompt is delivered to the task execution module (See sequence number 1). Depending on the identified task type, the system executes one of five different activities: generation, modification, retrieval, deletion, or clarification. For generation tasks, the task execution module generates a design result in JSON format and delivers it to the design implementation module, which then returns it to the NADIA interface (See sequence number 2). The 'Create()' methods for each building component class were used to enter the properties represented within the JSON-formatted design results and implement them into the BIM environment. For modification tasks, the task execution module initially requests the specification of the target. This specification is based on the building component types, referred to as

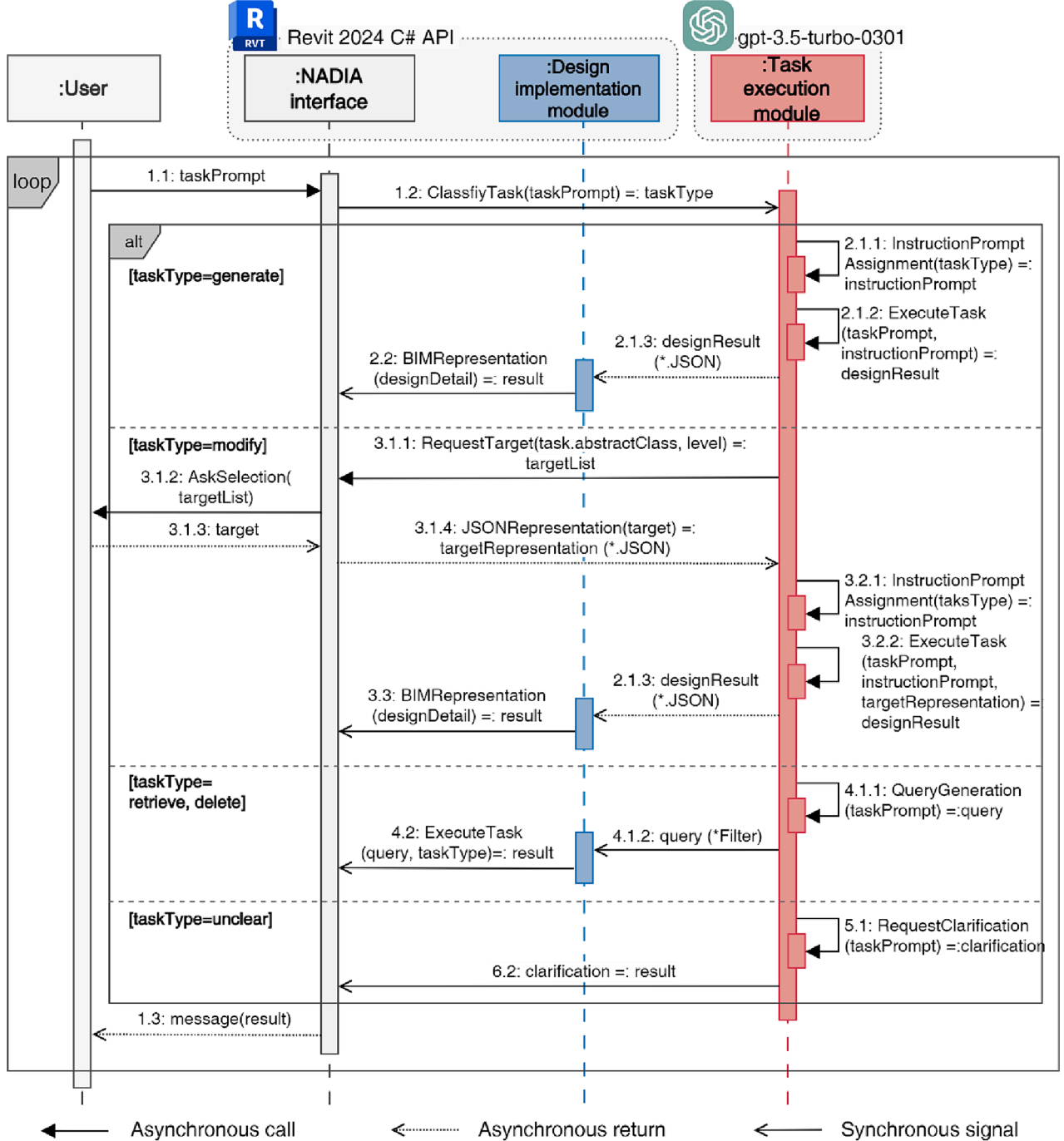


Fig. 6. UML sequence diagram of NADIA framework.

'abstract class,' and the level, defined by either 'subclass' or 'instance,' as identified by the task type (See sequence number 3). Once the target is specified, the NADIA interface retrieves it in a JSON representation and forwards it to the task execution module, along with the task and instruction prompts. To retrieve the target detail or object based on the task type, 'FilteredElementCollector' from the Revit API is utilized [62]. The resultant design is then delivered to the design implementation module and returned to the NADIA interface. To apply the changes, the corresponding 'Properties' of the target instances were modified. In cases where the task type is unclear, NADIA may request clarification (See sequence number 4). As shown in the UML sequence diagram, the task prompt and the selection of the target are the only types of input that the user provides to NADIA.

Fig. 7 is a screen capture of the NADIA interface. NADIA is implemented as an add-on that can be executed as an external command. Once activated, the interface module appears on the right side of the Revit environment as a separate panel, developed using the Windows Presentation Foundation (WPF). This panel comprises two text boxes and one button. The upper text box displays the history of the conversation, the lower text box allows users to type input task prompts, and the 'Send' button delivers the task prompt to the task execution and design implementation modules. The task prompt delivered to the modules will be displayed on the upper text box with a prefixed notation of '[User]'. Once the processing of the task prompt is completed, the upper text box will display the response of the modules with the notation '[NADIA]'. Once the execution is complete, NADIA may apply the results within the Revit environment. If the task was to generate or modify the design details, the results can be referred to within the properties windows upon selection of the relevant category. To provide a practical

understanding of the operation of NADIA within the BIM software, a video demonstration is included in [supplemental files as Video 1](#).

4. Validation

As previously outlined, the generation of exterior wall details was selected as the validation scope to evaluate NADIA's capability in executing design detailing in line with architects' requirements, considering its multifaceted nature. The validation specifically targeted the generation of a standard Revit wall detail. This study devised two distinct sets of task prompts to generate wall details under varying conditions and then evaluated the walls generated by NADIA for their adherence to the specified conditions and compliance with general engineering standards. The first set aimed to evaluate NADIA's performance to generate outputs that adhere to the given design conditions, thereby acting as a "design assistant." The second set aimed to demonstrate NADIA's capability in interpreting given conditions and proposing design details that comply with engineering performance standards, specifically thermal performance as required by the American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) 20.1–2019, thus acting as a "design consultant." The specific research questions for each role were:

- 1) How effectively does NADIA, as a design assistant, generate exterior wall details that adhere to the design requirements specified by an architect?
- 2) How effectively does NADIA, as a design consultant, generate exterior wall details that comply with thermal engineering requirements, with or without specific instructions for guiding references?

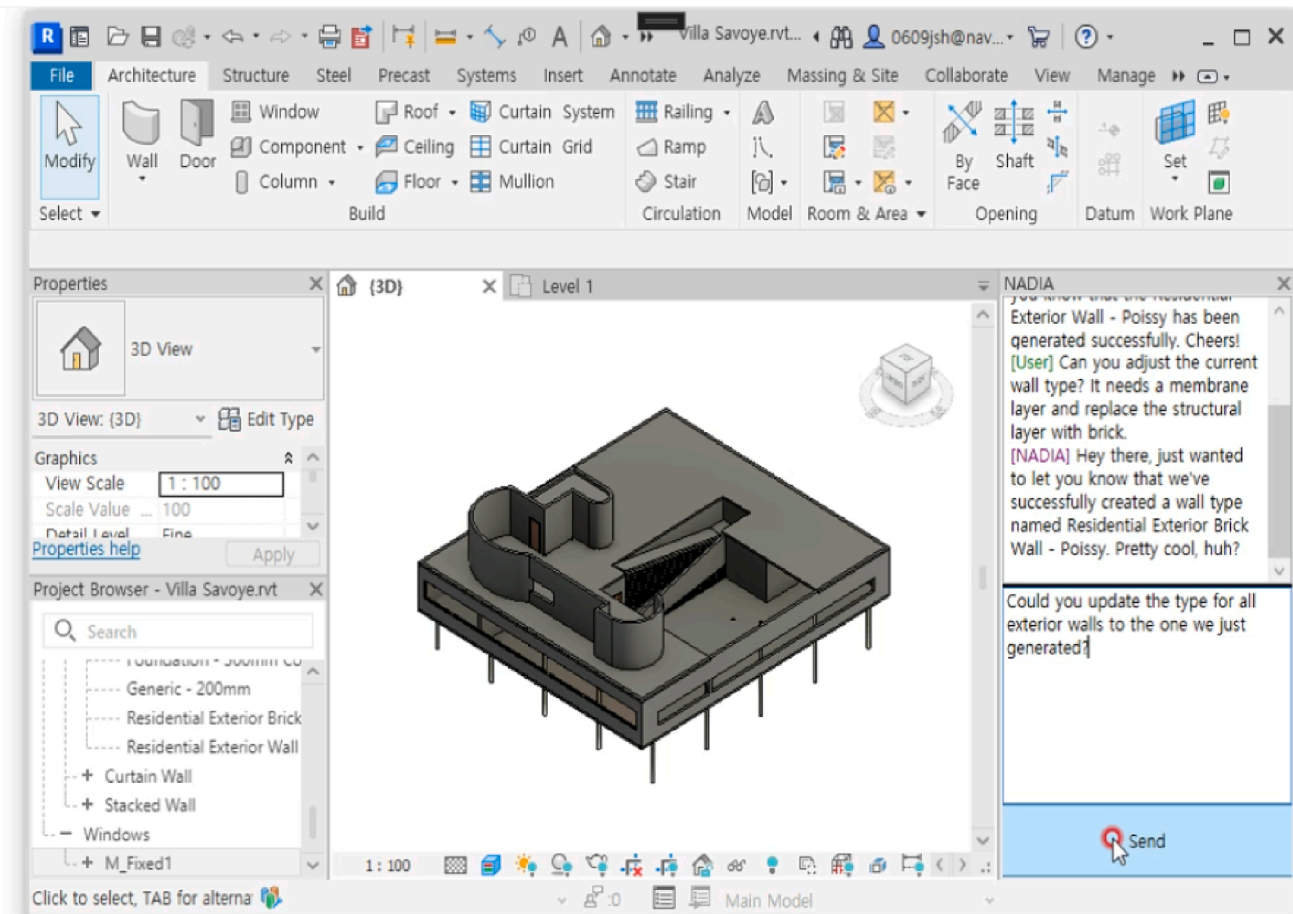


Fig. 7. NADIA interface.

The subsequent sections detail how each set of task prompts was designed to address the research questions.

4.1. NADIA as a design assistant

The first set of task prompts was designed to cover three conditions: the use of structural frame materials, insulation methods, and minimum structural thickness, with each chosen to evaluate the capability of NADIA in handling categorical, sequential, and dimensional properties.

The task prompts were designed by combining options from each condition, including ‘reinforced concrete’ or ‘timber’ as structural frame materials, ‘exterior insulation’ or ‘interior insulation’ as insulation method, and ‘140 mm’ or ‘190 mm’ for reinforced concrete, and ‘140 mm’ or ‘184 mm’ for timber walls, as illustrated in Fig. 8A. Each unique prompt code comprises the abbreviations of each condition: structural frame materials (C: reinforced concrete or T: timber), insulation methods (E: exterior insulation or I: interior insulation), and minimum structural thicknesses (1: 140 mm or 2: 190 mm for reinforced concrete walls and 1: 140 mm or 2: 184 mm for timber walls), as depicted in Fig. 8B. For example, CE1 represents a case with a reinforced concrete structure, exterior insulation method, and a minimum thickness of 140 mm. The corresponding task prompt for CE1 was “Propose a wall detail using a *reinforced concrete* structure and *exterior insulation* method, ensuring that the *reinforced concrete* has a minimum thickness of *140 mm*?” The italicized conditions were altered according to the prompt code for each of the eight unique task prompts. Each task prompt was input into the implemented NADIA system and executed 30 times to obtain statistically significant results by the central limit theorem, cumulatively generating 240 exterior wall details in total. The evaluation focused on determining the adherence of generated details to the specified conditions, employing binary criteria to manually distinguish between the “satisfied” and “not satisfied”:

- **Structural frame materials:** the material used in the structural layer of the generated wall detail was examined.
- **Insulation methods:** the sequence of insulation and structural layers within the wall details was examined.
- **Minimum structural thickness:** the thickness of the wall details’ structural layer was examined.

The performance of NADIA, both individually and collectively, in

satisfying the given conditions was thoroughly evaluated.

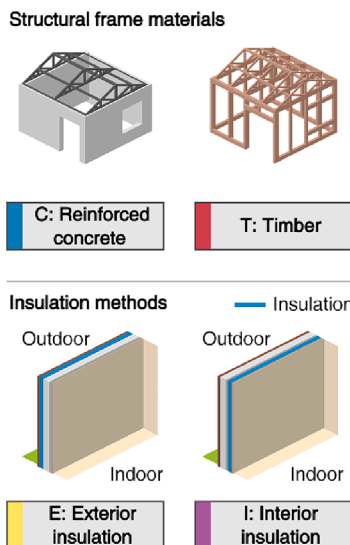
4.2. NADIA as a design consultant

The second set of task prompts was designed to evaluate whether the wall details generated by NADIA comply with general engineering standards, with a particular focus on their thermal performance. The thermal performance of the generated details was evaluated against the U-value (i.e., thermal transmittance value) requirements specified in the ASHRAE 90.1–2019 standard. This globally recognized standard not only establishes thermal performance criteria for buildings but also considers their cost-effectiveness, ensuring that the energy efficiency measures are economically feasible [63]. Specifically, ASHRAE 90.1–2019 details the thermal performance requirements for building envelopes in terms of U-values. The U-value specifications under ASHRAE 90.1–2019 vary based on factors such as structural frame materials, types of building occupancy, and climate zones.

The second set of task prompts was designed to reflect these factors as conditions, including ‘reinforced concrete’ or ‘timber’ as structural frame materials, ‘residential’ or ‘non-residential’ as building occupancy, and ‘very hot,’ ‘warm,’ ‘cool,’ or ‘very cold’ as climate zones (Fig. 9A). Each unique prompt code comprises the abbreviations of each condition: structural frame materials (C: reinforced concrete or T: timber), building occupancy types (R: residential or N: non-residential), and climate zones (1: very hot, 3: warm, 5: cool, or 7: very cold), as depicted in Fig. 9B. For example, CR1 represents a case with a reinforced concrete structure and a residential project located in a very hot climate. The corresponding task prompt written for CR1 was, “Propose a wall detail using a *reinforced concrete* structure for a *residential* project located in a *very hot* climate.” Similar to the previous validation method, the italicized conditions were altered according to the prompt code.

Additionally, this set of task prompts included an ablation study to test the effect of varying levels of specificity for reference engineering criteria: no reference information; only the reference name (e.g., ASHRAE); the reference number (e.g., ASHRAE 90.1–2019); and specific ASHRAE U-value requirements are included in the task prompts (Fig. 9C). For example, to provide thermal insulation requirements with a specific reference name and number, the phrase “while conforming to ASHRAE 90.1–2019” was added at the end of the task prompt. To provide thermal insulation requirements in the form of specific U-values for different conditions, the U-value table was given in a comma-separated

A Conditions



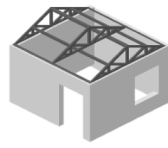
B Task prompts

Prompt code	Structural frame materials	Insulation methods	Minimum thicknesses
CE1	C	E	1
CE2	C	E	2
CI1	C	I	1
CI2	C	I	2
TE1	T	E	1
TE2	T	E	2
TI1	T	I	1
TI2	T	I	2

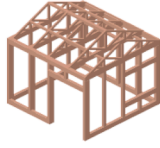
Fig. 8. Design for the first task prompt set.

A Conditions

Structural frame materials



C: Reinforced concrete

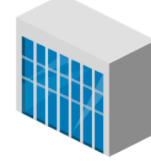


T: Timber

Building occupancy types



R: Residential



N: Non-residential

Climate zones

1: Very hot

3: Warm

5: Cool

7: Very cold

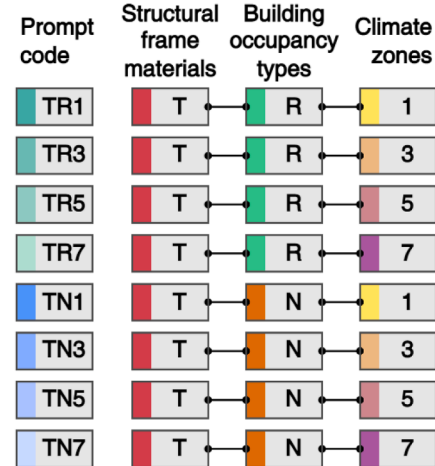
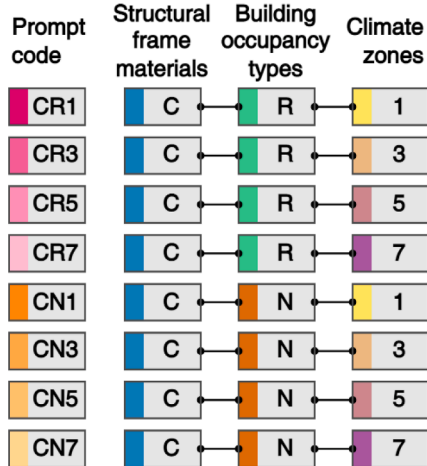
2: Hot

4: Moderate

6: Cold

8: Subarctic

B Task prompts



C Reference information

- No reference: ""
- Reference name: ", while conforming to ASHRAE"
- Reference name and number: ", while conforming to ASHRAE 90.1-2019"
- Engineering requirements in a textual format: ", while conforming to given exterior wall U-value ($\text{W/m}^2\cdot\text{K}$) criteria"

Building occupancy types	Structural frame materials	Climate zones							
		Very hot climate	Hot climate	Warm climate	Mixed climate	Cool climate	Cold climate	Very cold climate	Subarctic climate
Nonresidential	Mass	3.293	0.857	0.698	0.591	0.511	0.454	0.403	0.273
Nonresidential	Metal building	0.534	0.534	0.534	0.341	0.284	0.284	0.250	0.221
Nonresidential	Steel-framed	0.704	0.477	0.437	0.363	0.312	0.278	0.278	0.210
Nonresidential	Wooden-framed	0.505	0.505	0.505	0.363	0.290	0.290	0.290	0.182
Residential	Mass	0.857	0.698	0.591	0.511	0.454	0.403	0.403	0.273
Residential	Metal building	0.534	0.534	0.409	0.284	0.284	0.284	0.250	0.210
Residential	Steel-framed	0.704	0.363	0.363	0.363	0.312	0.278	0.238	0.210
Residential	Wooden-framed	0.505	0.505	0.363	0.363	0.290	0.290	0.290	0.182
Semiheated	Mass	3.293	3.293	3.293	3.293	0.857	0.857	0.698	0.591
Semiheated	Metal building	1.999	0.920	0.920	0.920	0.534	0.534	0.409	0.341
Semiheated	Steel-framed	1.999	0.704	0.704	0.704	0.477	0.477	0.363	0.363
Semiheated	Wooden-framed	1.658	0.505	0.505	0.505	0.505	0.505	0.363	0.290

Fig. 9. Design for the second task prompt set.

value (CSV) format after adding the phrase “while conforming to the given exterior wall U-value ($\text{W}/\text{m}^2\cdot\text{K}$) criteria” at the end of the task prompt.

Overall, 1,920 wall details were generated from 30 executions of 64 distinct prompts, comprising 16 unique task prompts across 4 distinct levels of specificity for reference engineering criteria. These wall details were analyzed to evaluate whether the thermal performance of the NADIA-generated wall details satisfies the ASHRAE 90.1–2019. The total U-value of wall detail was calculated from the sum of its components’ thermal resistances, calculated using the equation:

$$U_{\text{total}} = \frac{1}{\sum_{i=1}^n R_i}$$

where U_{total} denotes the overall U-value, n denotes the number of wall detail layers, and R_i denotes the thermal resistance of the i^{th} individual layer, calculated as the thickness of the wall layer divided by its thermal conductivity. The U-values of each generated detail were calculated and evaluated to determine if they exceeded (i.e., underperformed) the U-value requirements specified under ASHRAE 90.1–2019.

All the task prompts included in the two sets, along with the wall

details generated from the executions, are included in the [Supplemental files as Data 1 and Data 2](#), respectively.

5. Results

5.1. NADIA as a design assistant

The performance of NADIA as a design assistant was analyzed based on how well the NADIA-generated wall details satisfied the following requirements: type of structural frame material, minimum thickness of the structural layer, and type of insulation method.

5.1.1. Satisfying structural frame materials

Fig. 10 illustrates a three-layer Sankey diagram showing the distribution of the structural frame materials. The first layer denotes the structural frame options—reinforced concrete or timber frame—applied to the task prompt used to generate the design details. The second layer denotes the actual structural frame type generated, and the last layer denotes the individual material names used for the structural layer. Among the 120 wall details generated when the use of reinforced

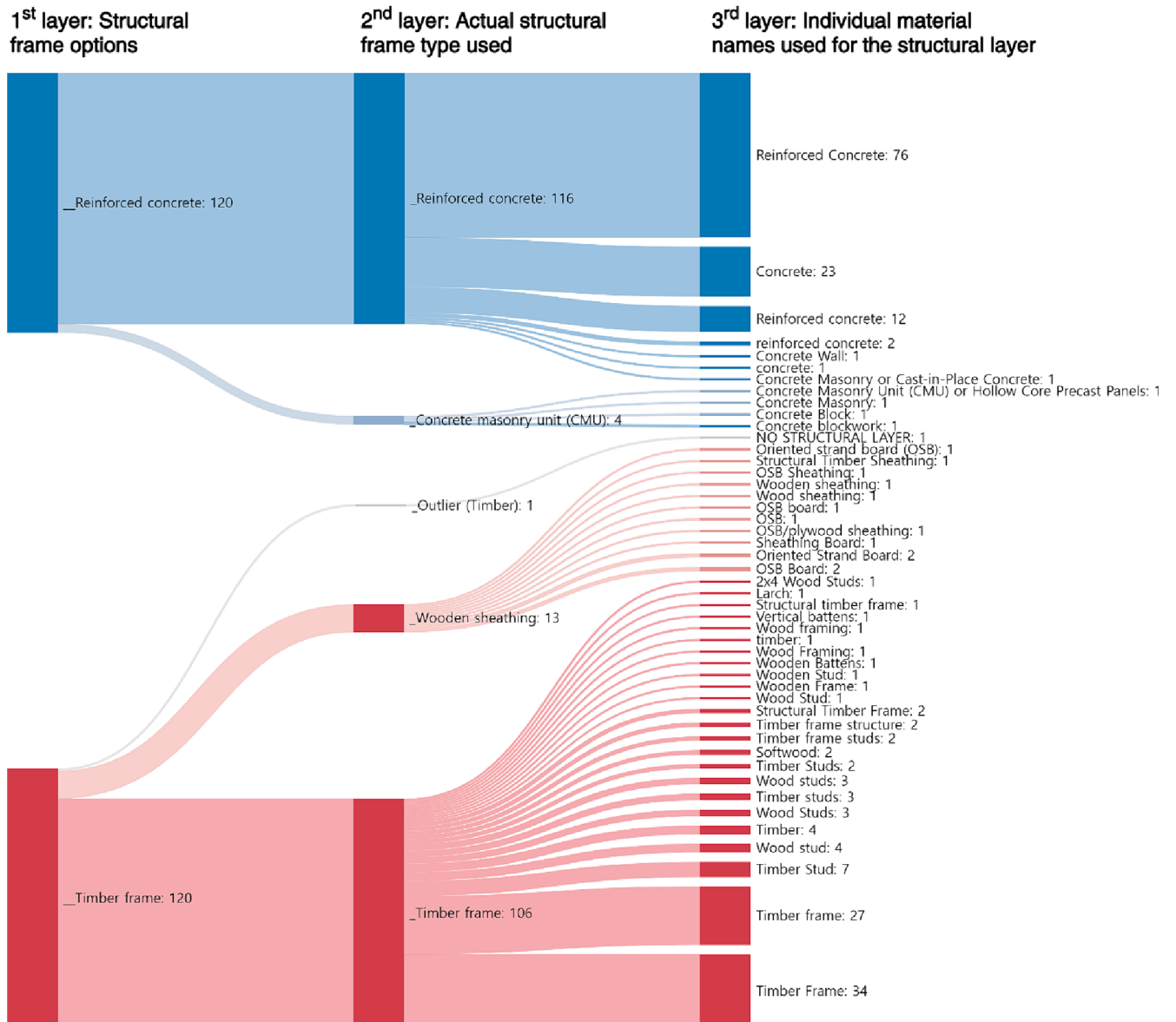


Fig. 10. Distribution of structural frame materials.

concrete was required, 116 accurately reflected the request to generate a mass concrete wall, while four details used concrete masonry units (CMU), as presented at the top of Fig. 10. Meanwhile, among the 120 wall details generated when the use of timber-framed structure was required, 106 were accurate, 13 were wooden sheathing details without a timber frame, and one had neither a timber frame nor substrate, as presented at the bottom of Fig. 10.

5.1.2. Satisfying structural layer thicknesses

Fig. 11 is a box plot that depicts the distribution of the structural layers' thicknesses, with the dashed lines representing the minimum thickness values. The analysis of the minimum thickness condition involved examining instances where the structural layers were thinner than the specified minimum thickness. Among 120 wall details, 103 (indicated by a short solid line in each category in Fig. 11) satisfied the minimum thickness condition as required in the task prompt. The remaining 17 cases resulted in details whose structural layers were thinner than the requested minimum. These instances are represented as dots in the figure and include one instance each for CI2, TE1, and TI2; two for CE2; and six for TI1, and TI2.

5.1.3. Satisfying insulation methods

Fig. 12 illustrates a three-layer Sankey diagram showing the use of the insulation methods. The first layer denotes the insulation method options—exterior or interior insulation—applied to the task prompt used to generate the wall details. The second layer denotes the actual insulation method used within the detail, and the last layer denotes if the use of the insulation types corresponds to the given conditions.

Four different insulation methods were used within the wall details generated by NADIA: the double insulation method, where insulation layers were placed on both sides of the structural layer; the exterior insulation method, with insulation layers only on the exterior side; the interior insulation method, with insulation layers only on the interior side; and a case where the insulation layer existed without a structural layer.

Among the 120 wall details generated with the exterior insulation method, 110 instances included exterior insulation, whereas 10 did not. Among the 120 generated with the interior insulation method, 106 of those included interior insulation, 13 instances did not include interior insulation and a single instance was comprised of an insulation layer without a structural or substrate layer. Among the 240 wall details, 145 were generated using the double-insulation method, which is a common practice used for placing insulation materials.

5.1.4. Overall performance in satisfying design requirements

Fig. 13 illustrates the performance of NADIA as a design assistant across different criteria: overall conditions (Fig. 13A), structural frame material (Fig. 13B), insulation method (Fig. 13C), and minimum structural thickness (Fig. 123) requirements. NADIA's mean accuracy on overall conditions was 83.33 %, with a standard deviation of 11.18 %. It

showed higher accuracy in tasks involving reinforced concrete compared to those with timber frames and interior insulation methods, both for structural frame materials and minimum structural layer thickness. Specifically, accuracies for reinforced concrete were 96.67 % and 97.25 % for structural frame materials and minimum structural thickness, respectively. For timber frames, these accuracies were 88.33 % and 87.5 %. There was also a marked discrepancy between exterior and interior insulation methods with accuracy of 97.50 % and 85.83 %, respectively.

Although NADIA generally performed better with certain requirements like reinforced concrete walls and exterior insulation methods, there is room for improvement, especially in other areas like timber frames and interior insulation. Overall, NADIA demonstrated a solid average accuracy in producing exterior wall details that satisfy specific design conditions required by the task prompt.

5.2. NADIA as a design consultant

Fig. 14 presents box-and-whisker plots of U-value distributions under different conditions: no reference (Fig. 13A); the reference name (i.e., ASHRAE) (Fig. 13B); the reference name and code numbers (i.e., ASHRAE 90.1–2019) (Fig. 13C); and the textual engineering requirements (Fig. 13D). The inverted triangles of each box plot indicates the maximum U-value requirements required by ASHRAE 90.1–2019, with values below these triangles considered satisfactory.

Out of the 1,920 wall details, 1,894 instances met the U-value requirements. There were 16 cases in which the U-value of the wall detail existed outside of the box-and-whisker plots, and thus are considered outliers. Among the 64 sets of resulting box-and-whiskers, four cases—CR3 and TR5 in Fig. 14A, CR3 in Fig. 14B, and TN5 in Fig. 14C—exceeded the requirements values, including ten cases that did not satisfy the requirements. The placement of box-and-whiskers and outliers in the analysis indicates that the U-values of wall details generated by NADIA generally satisfy ASHRAE thermal comfort standards, showcasing its effectiveness.

The accuracy of the NADIA-generated wall details in satisfying the U-value requirements was analyzed across different levels of specificity as depicted in Fig. 15. The accuracy varied between 86.67 % to 100 %, with the highest mean accuracy of 98.96 % being achieved when only the reference name was provided. This was followed by 98.75 % with specific engineering values, 98.54 % without any reference, and 98.33 % when both reference names and numbers were provided. The accuracy increased slightly when the reference name was provided (by 0.42 %) and when engineering requirements were given in textual format (by 0.21 %). However, the accuracy decreased slightly (by 0.21 %) when both the reference name and number were specified. In terms of accuracy, NADIA performed near-perfectly in all cases but the relationship between levels of specificity and the performance was not identified. The standard deviation in accuracy was 3.44 % for cases where no reference was given. Specifically, the standard deviation was lower at

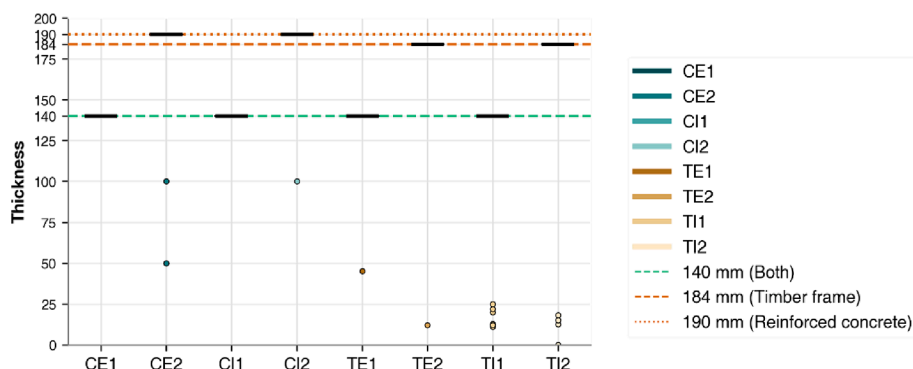


Fig. 11. Distribution of the thickness for each structural layer.

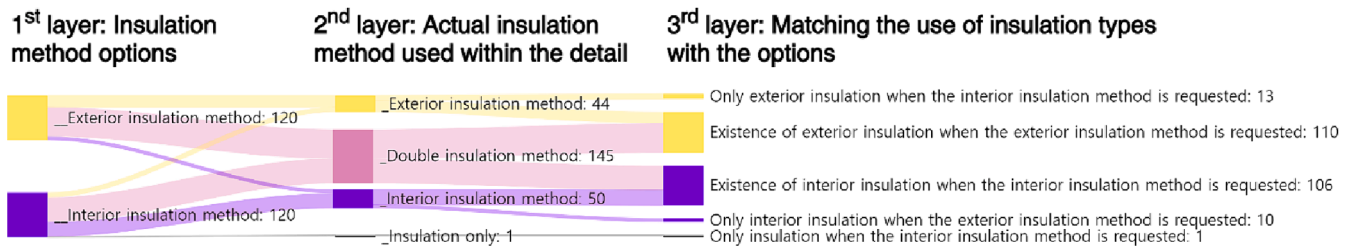
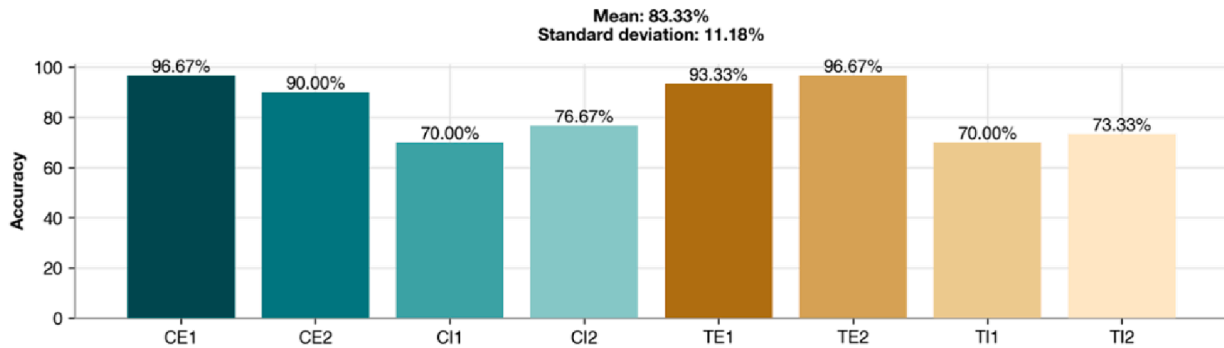
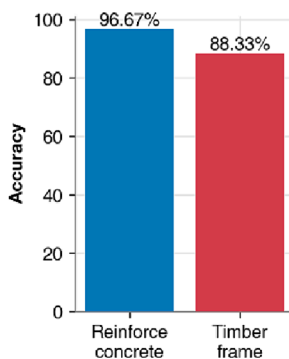


Fig. 12. Distribution of insulation methods.

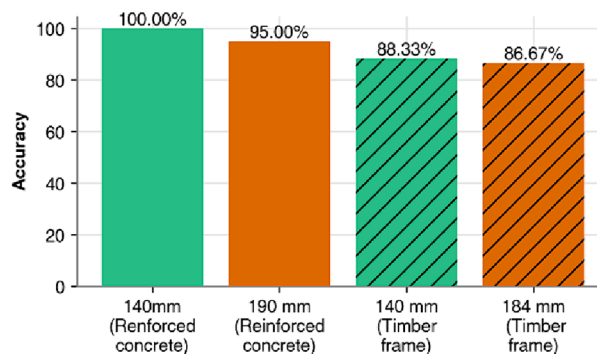
A Accuracy in satisfying all the requested conditions



B Accuracy in satisfying structural frame materials requirements



C Accuracy in satisfying minimum structural layer thicknesses requirements



D Accuracy in satisfying insulation methods requirements

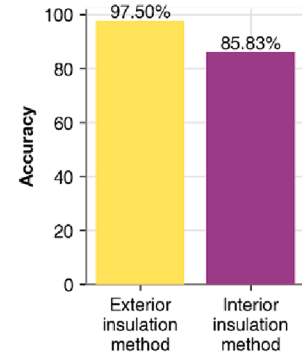


Fig. 13. Performance of NADIA as a design assistant.

2.35 % when only the reference name was provided, as opposed to 2.43 % and 2.06 % in cases where both reference names and numbers were specified. These results indicate that NADIA can generate wall details that comply with thermal engineering requirements while showing potential to provide a more consistent performance in conforming to the requirements when they are specified.

6. Discussion

This study quantitatively validated NADIA's ability to generate design details that adhere to user-specified design requirements and ungiven engineering requirements, while also identifying areas for improvement. Despite achieving notable accuracies—92.5 % for minimum thicknesses and 96.67 % for the application of structural frames in reinforced concrete structures—disparities were observed. Performance discrepancies in NADIA suggest gaps in comprehending specific design requirements, which may be attributed to an imbalanced training dataset or limitations in the representation of building components within an OOP data structure. Accuracy decreased to 88.33 % for timber

frame requests and to 85.83 % for interior insulation methods, potentially due to an insufficient training dataset or the inherent complexity of timber-framed walls compared to those of reinforced concrete [64].

In addition, NADIA's performance varied depending on its role as a design assistant and a design consultant: i.e., when its accuracy was evaluated based on its ability to generate details according to given design requirements (as a design assistant) versus its ability to satisfy unspecified engineering requirements with or without guiding references (as a design consultant). When evaluated as a design assistant, NADIA achieved a mean accuracy of 83.33 % in generating design details that adhered to specified dimensions or construction methods (i.e., prescriptive specifications). In contrast, as a design consultant, it excelled with a mean accuracy of 98.65 %, effectively generating design details that complied with general engineering requirements when provided with contextual information related to desired performance (i.e., performance-based specifications). This discrepancy between NADIA's performance as a design assistant versus a design consultant can be attributed to the inherent characteristics of LLMs. LLMs excel in generating viable solutions when provided with contextual information,

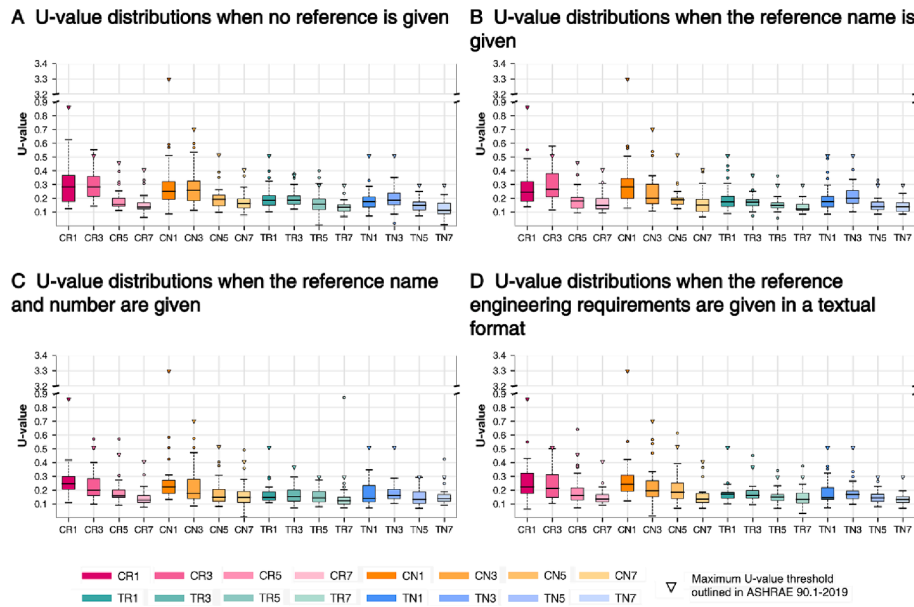


Fig. 14. Distributions of the U-values of the wall details generated through NADIA.

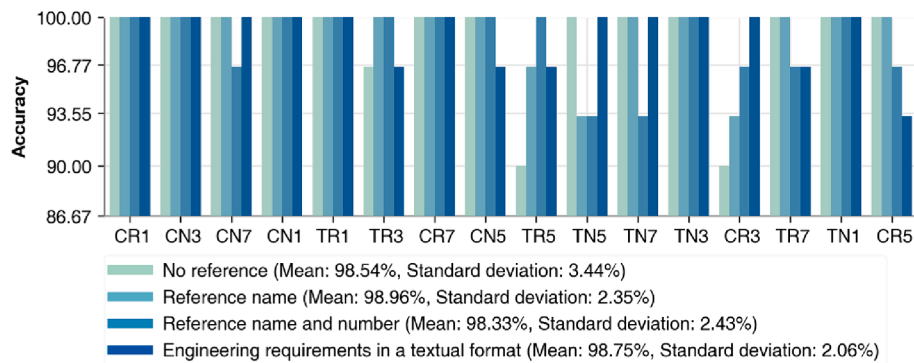


Fig. 15. Performance of NADIA in satisfying the thermal performance requirements of ASHRAE 90.1–2019.

leveraging their extensive knowledge base to infer and apply general engineering requirements creatively [49,65]. However, their performance in strictly adhering to specified conditions can be less consistent, potentially due to limitations in understanding the precise context or nuances of specific design requirements [54]. This underscores a critical area for improvement in enhancing NADIA's ability to accurately parse and apply detailed specifications.

On the other hand, the tendency of NADIA to exceed specified requirements was observed. For instance, as a design assistant, NADIA showed a tendency to adopt a double insulation method when the use of either interior or exterior insulation was requested, as shown in Fig. 12. Although double insulation is recognized as an ideal method to enhance insulation performance, its practical application is limited due to resource considerations. Similarly, when evaluated as a design consultant, NADIA demonstrated the capability to generate design details with thermal performance that complied with ASHRAE 90.1–2019 standards with very high accuracy (achieving a mean accuracy of 98.96 %). However, the median U-values of the wall details generated by NADIA were significantly lower than the specifications set by ASHRAE 90.1–2019. This may indicate a characteristic inclination of LLMs to provide the “safest answer” in order to minimize risk or error [49], which can be impractical in some cases. This observation highlights potential opportunities for linking design optimization modules to NADIA to iteratively improve automatically generated details before NADIA returns the outcomes. To address these challenges, we propose

the following solutions:

- **Fine-tuning:** Enhance the diversity of component types and include specific detail types in the training dataset for fine-tuning, aiming to improve the comprehension and representation of BIM objects.
- **Retrieval-augmented generation (RAG):** Implementing RAG techniques can boost domain-specific performance by utilizing detailed, context-specific knowledge from reference documents [66]. This approach allows for more controlled outcomes based on factual references.
- **Enhancing contextual understanding and optimizing design solutions:** Future efforts should aim at improving NADIA's comprehension of the context and topology within BIM models, essential for accurately interpreting and adapting to a broader spectrum of building components and tailored details. Simultaneously, integrating multi-objective optimization tasks and/or design quality checking will enable NADIA to balance various design criteria, facilitating the creation of more practical and efficient design solutions. These approaches will significantly advance NADIA's utility in generating refined and optimized architectural designs.

Advancements in these areas are essential for developing NADIA into a more effective and efficient tool for architectural design detailing, ensuring it not only handles a broader scope of design authoring but

further optimizes user-specified and engineering requirements.

7. Conclusion

The complexity of learning a BIM authoring tool has posed a challenge, often hindering the widespread adoption of BIM. This paper proposed the NADIA framework which integrates LLM and BIM authoring tools by employing LLM chaining to provide an interactive design detailing environment using natural language as the main medium. The framework features an interface module integrated within the BIM environment, which accepts textual inputs from users. These inputs are processed by an LLM-based task execution module that employs LLM chaining, dividing the design detailing process into text classification and generation tasks which are guided by the instruction prompts. The interim results generated using the task execution module are then seamlessly delivered within the BIM environment through a design implementation module. The performance of the NADIA system, implemented using GPT-3.5 and Revit APIs, was validated by focusing on the generation of exterior wall details. The validation aimed to evaluate if NADIA could generate wall details that adhere to specific design conditions while complying with engineering requirements. As a design assistant, NADIA achieved an average accuracy of 83.33 % in satisfying multiple design conditions, including the use of structural frame materials, adherence to their minimum structural thickness requirements, and the selection of insulation methods. As a design consultant, NADIA attained an average accuracy of 98.65 % in complying with the thermal engineering requirements as outlined in ASHRAE 90.1–2019. Although there is potential for further improvement, this study highlights the capabilities of the NADIA framework in assisting architects to effectively communicate with AI and develop design details using natural language.

The contributions of this study are as follows. First, this study proposes a specific LLM and BIM chaining framework, named NADIA, for BIM-based detailing (Fig. 1) through three key ideas. While a potential use of LLM for BIM detailing may have been considered, a specific method has not been proposed. The text describes three main tasks: 1) formalizing the interactive design processes among an architect, an LLM, and a BIM authoring tool by separating a detail specification task and a detail generation task (Fig. 2); 2) developing an instructional prompt template (Fig. 3, Fig. 4, and Fig. 5) to direct an LLM to produce “engineering-wise rational” details in a machine-readable format while minimizing the creation of nonsensical designs; and 3) proposing a method (Fig. 6) to chain an LLM and a BIM authoring tool. Second, through the validation tests, it identified the strengths and weaknesses of an LLM-based BIM detailing as a design assistant and a design consultant, which provides directions leverageable for future developments in LLM-based design automation research.

While this study demonstrated that NADIA can generate exterior wall details satisfying the requirements in terms of the design requirements and engineering requirements (thermal performance), further work is required for its use in practical applications. First, the current framework relies primarily on a limited set of instruction prompts and thus requires the expansion of the set to diversify the supported components beyond exterior walls, such as columns, slabs, or doors. Secondly, despite NADIA’s high performance, there is still room for improvements in the prompts to make NADIA recognize the context and topology of a target design. Thirdly, more advanced LLM-based decision-making methodologies such as chain of thought [67], or ensemble tree-based methods (i.e., tree of thoughts) [68], could be explored to enhance the performance of LLM. Finally, NADIA cannot generate new object classes, and therefore cannot generate instances of non-existing classes. Filling this gap would significantly increase the framework’s adaptability and usefulness, enabling full integration into the practical BIM authoring process.

This research marks a preliminary step in chaining LLMs with BIM systems. Future endeavors will build on the insights and limitations

identified, aiming to broaden NADIA’s capability to encompass a wider array of design tasks in the BIM environment. Real-world trials and enhancing performances by adopting fine-tuning and RAG are also planned to further refine the framework. Progressing this technology is in line with the visions set forth by Douglas Engelbart, Steven Coons, Ivan Sutherland, and Charles Eastman, who envisioned a future where designers and intelligent machines collaborate seamlessly in the creative process.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used OpenAI’s tool, ChatGPT in order to proofread the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRedit authorship contribution statement

Suhyung Jang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Ghang Lee:** Writing – review & editing, Funding acquisition, Conceptualization. **Jiseok Oh:** Writing – review & editing, Validation. **Junghun Lee:** Validation, Writing – review & editing. **Bonsang Koo:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data generated from the experiment of this study is attached as [Supplement files](#).

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.aei.2024.102532>.

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